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1.0 INTRODUCTION:

At the request of **Mr. Shola Michael** and **Mrs. Elizabeth Yetunde**, occasioned by their s' advice and upon receiving payments, Subsoil Investigations has been carried out at above named site and we hereby present our findings upon completion of all fields & laboratory works. The objective of these investigations is to determine relevant Geotechnical/Engineering parameters of the subsoil strata that underlay the site i.e. subsurface soil condition, groundwater condition, earthwork and foundation design and construction of the site Civil works, which will aid economical design of foundation(s) for the proposed **Development**.

2.0 SCOPE OF INVESTIGATIONS:

The investigations involved soil boring using a Pilcon Wayfarer Shell and Auger cable percussion-drilling rig, Penetration Testing using 2.5-Tones Dutch Cone Penetration Testing (DCPT) Machine, Visual assessment of the site/description of subsoil encountered and laboratory testing's of selected subsoil and water samples recovered from the Boreholes.

One Borehole (1 BH) and Two Cone Penetration tests (2nos. CPT) shown on the Site Plan Figure 1 were carried out as agreed/presented to us prior to commencement of field works.

2.1 Site Work & Methodology:

One number (1no.) soil test Borehole using Shell and Auger cable percussion rig was drilled up to 10.0m below the existing ground level upon encountering Medium dense to dense sand and also 2nos. Cone



During boring, samples were recovered at regular intervals of 0.75m, while standard penetration tests (SPT) were carried out at alternative intervals of 1.5m.

General notes on the Shell and Auger boring method are contained in Appendix of this report. The results of the Borehole are presented as Borehole figure 2.

The Cone Penetration tests (CPT) were executed using the 2.5-Tones capacity machine. The test measures the relative in-situ density of the soil over the depths tested. The results of this test is presented as CPTs -1 to 2.

General notes and more details on the Cone Penetration tests are presented in Appendix of this report.

Depths referred to in this report are approximate and occur below the existing ground level as at the time of investigations.

2.2. Laboratory Analyses:

Samples were retrieved from the Borehole in the course of drilling at different depths. Both Disturbed and Undisturbed samples were recovered in accordance with British Standards Institutes (B.S 5930: Code of practice for site investigations).

The following laboratory analyses were carried out on selected samples recovered from the boreholes using the procedures specified in the British Standard B.S 1377: 2015. Methods of Testing of Subsoil for Civil Engineering Purposes”.

- (i) 4nos. Sieve Analysis for classification purposes;
- (ii) 4nos. Bulk density for classification purposes;